MachineSense

Installation and User Guide

NUS-ISS Pattern Recognition Systems Practice Module · Group 01 · Bhuvesh Kumar (A0243823H) · October 2026
Repository: github.com/bhuveshsharma09/MachineSense

MachineSense diagnoses faults in a rotating machine from one five-second recording. It answers three questions: is the machine healthy (Q1), which of ten known conditions it resembles (Q2), and whether it may have a fault the models have not seen (Q3). The web app shows each answer with its measured evidence.

This guide explains how to install the system, start it, and use each page. You do not need the MAFAULDA dataset. The submission includes the trained models and five demo recordings.

> **Research demonstration.** MachineSense was trained and tested on one bench-top rig with induced faults. It is not validated on real machines. A fault name is a model prediction, not a confirmed cause.

Contents

1. What is included
2. Requirements
3. Install
4. Start the system
5. Use the app
6. Optional: written explanations and the assistant
7. Troubleshooting

1. What is included

Folder	Contents
`SystemCode/`	All code: the `ml` scoring package, the FastAPI backend, the React frontend, notebooks, scripts and tests
`SystemCode/artifacts/bundle/`	The trained models (CNN-6, AE-6, B1s, Mahalanobis familiarity, logistic regression), thresholds and decision rule, about 4 MB, with SHA-256 hashes
`SystemCode/demo/`	Five demo recordings and four healthy reference recordings
`SystemCode/HOW_TO_RUN.md`	The same quick start in text form, and the full reproduction steps

2. Requirements

Item	Version or size
Python	3.12
Node.js (with npm)	22
Git	Any recent version (only to clone)
Free disk space	About 3 GB for the Python packages
Processor	Any CPU. No GPU is needed. A diagnosis takes about 1 second.
Operating system	macOS, Linux or Windows

3. Install

Step 1. Get the code. Either unzip the submission, or clone the repository:

```
git clone https://github.com/bhuveshsharma09/MachineSense.git
cd MachineSense/SystemCode
```

Step 2. Create a Python environment and install the packages. This takes about 10 minutes, mostly downloads.

```
python3.12 -m venv .venv
source .venv/bin/activate            # Windows: .venv\Scripts\activate
pip install --upgrade pip
pip install -r requirements.txt
pip install -e .
```

Linux or Windows: install the CPU-only PyTorch *before* `requirements.txt`. Otherwise pip downloads several gigabytes of GPU packages.

```
pip install torch==2.14.0 torchvision==0.29.0 --index-url
https://download.pytorch.org/whl/cpu
```

macOS: `brew install libomp` is needed only for the LightGBM experiments. The app itself does not use LightGBM.

4. Start the system

Step 3. Start the backend. In `SystemCode/`, with the environment active:

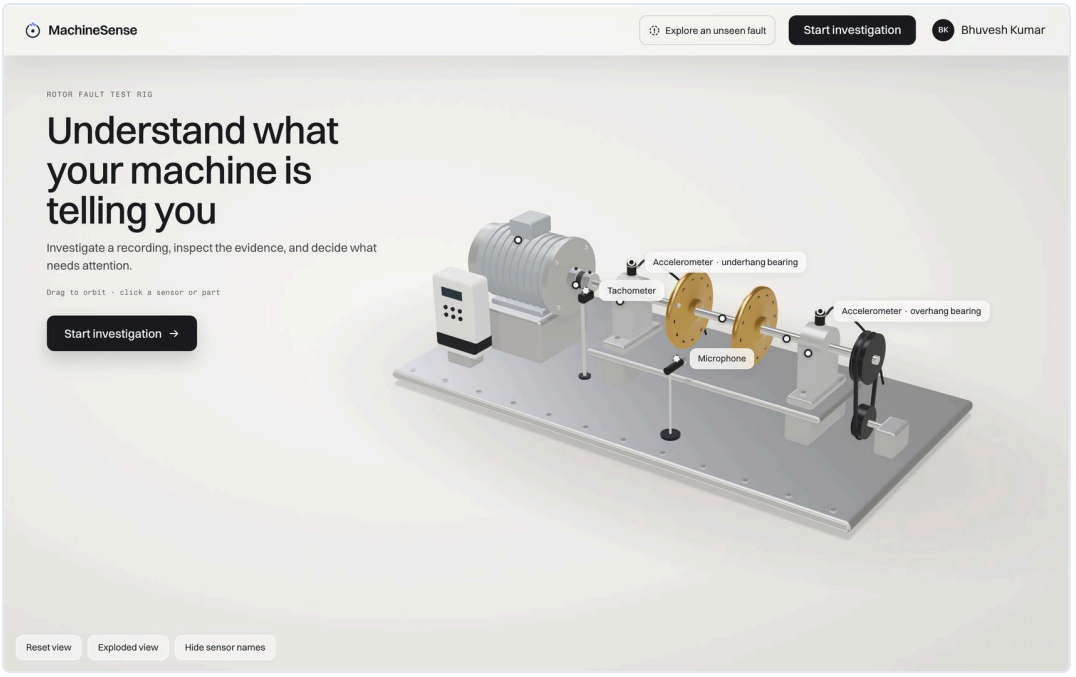
```
uvicorn backend.app:create_app --factory --host 127.0.0.1 --port 8000
```

At startup the backend checks the SHA-256 hash of every model file. It refuses to start if one does not match. To confirm it is ready, open `http://127.0.0.1:8000/api/health`. You should see `"bundle_verified": true` and `"mode": "demo-pack"`.

Step 4. Start the frontend in a second terminal:

```
cd SystemCode/frontend
npm ci
npm run dev
```

Step 5. Open `http://localhost:5173` in a browser. Enter any display name. No account is needed.



The home page, with the 3D model of the test rig and its sensors.

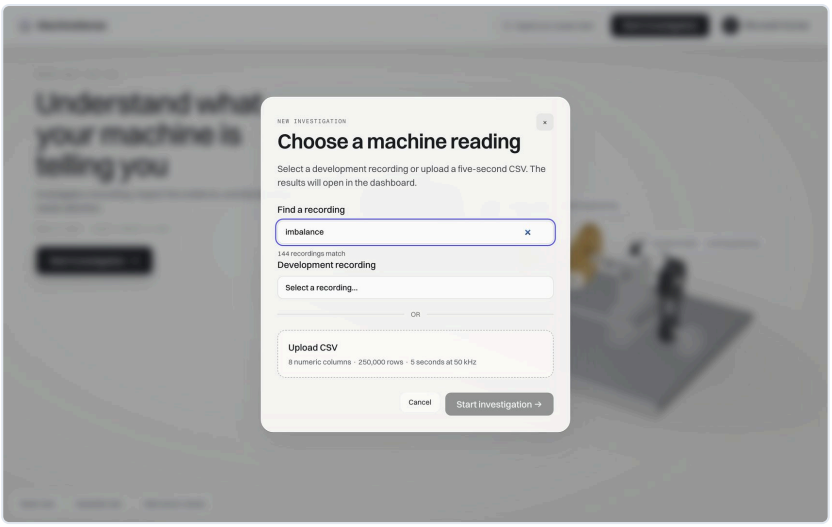
5. Use the app

5.1 Start an investigation

Choose **Start investigation**. The dialog lists the five demo recordings. Pick one, or upload your own recording (Section 5.7).

Recording	What the app shows
`normal__24.576`	Healthy
`imbalance__30g__19.456`	Known fault: imbalance
`horizontal-misalignment__1.0mm__24.9856`	Known fault: horizontal misalignment
`underhang__cage_fault__6g__35.0208`	Known fault: underhang cage (see the envelope view)
`imbalance__10g__13.9264`	Further inspection needed

The demo recordings are development recordings from the validation split, which set the decision thresholds. Their verdicts are therefore in-sample and are not evidence of accuracy. The reserved test recordings are never shipped, listed or read by the app.



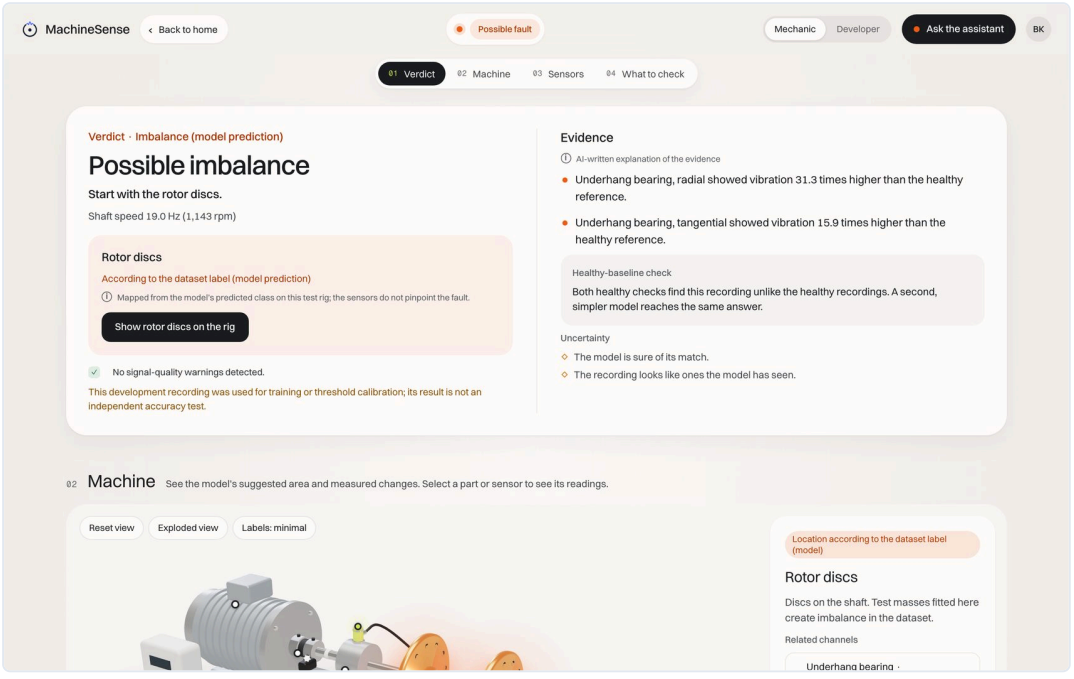
The Start investigation dialog.

5.2 Read the verdict (Mechanic view)

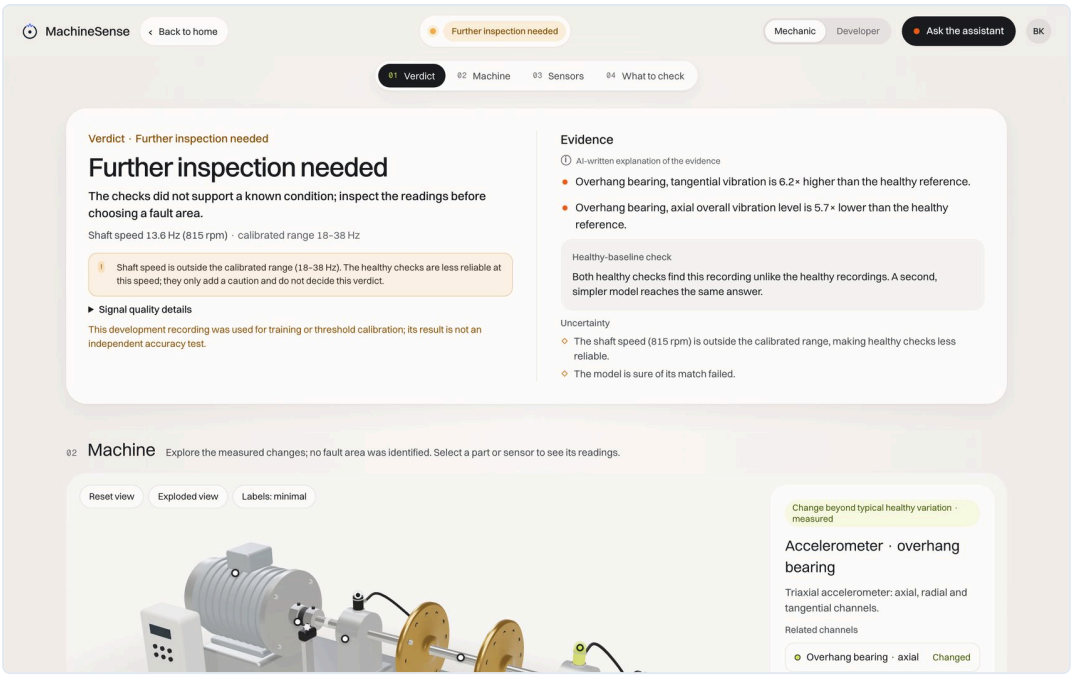
The Mechanic view is the default. It shows one of three verdicts:

Verdict	Meaning
Healthy	The recording matches healthy behaviour. A caution is added when a healthy-only check objects.
Possible fault (model prediction)	The classifier is confident and the pattern is familiar. The location is given according to the dataset label.
Further inspection needed	The confidence or familiarity check failed. The system does not guess a label.

Below the verdict, the page shows the rig with the relevant part highlighted, the sensors that changed compared with a healthy recording at a similar speed, and suggested checks from the guidance library.



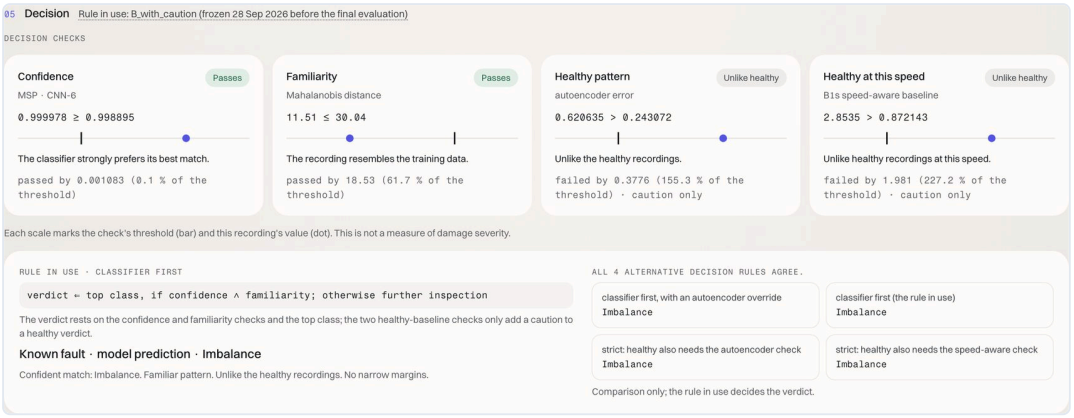
A known-fault verdict: imbalance, shown as a model prediction.



An uncertain case: further inspection needed.

5.3 Developer view

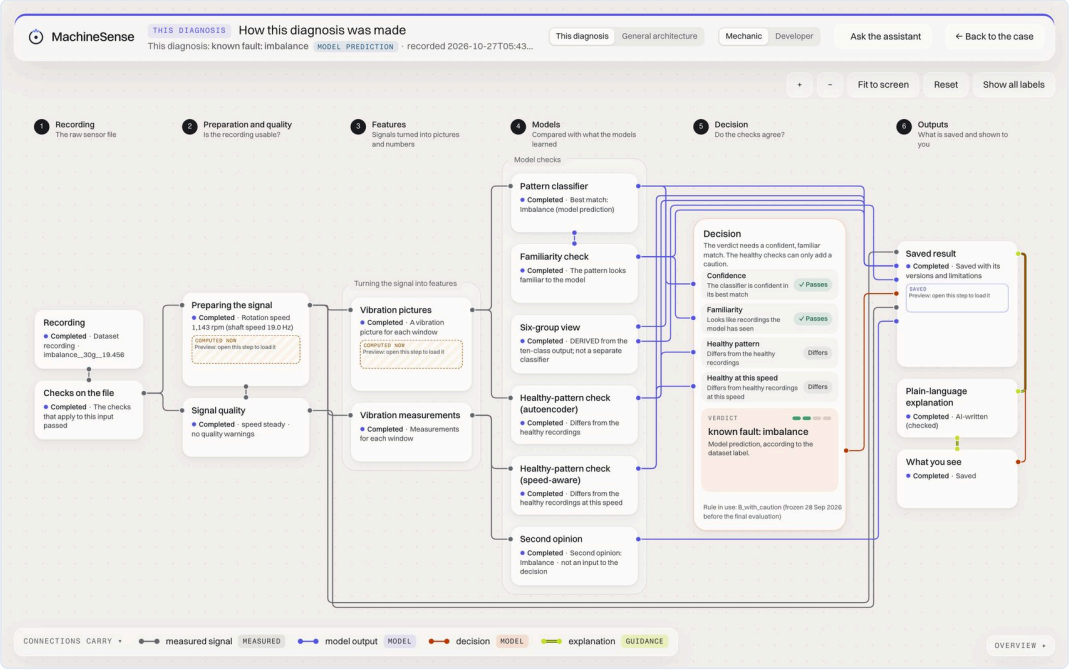
Switch to **Developer** at the top of the dashboard. It shows every model score against its threshold, the top classes, signal quality and why the decision rule chose the verdict.



The Developer view: the decision section.

5.4 Pipeline page

From a case, open **How this diagnosis was made**. The page shows the six stages of the diagnosis (recording, preparation and quality, features, models, decision, outputs) with the stored values and timing of each stage.



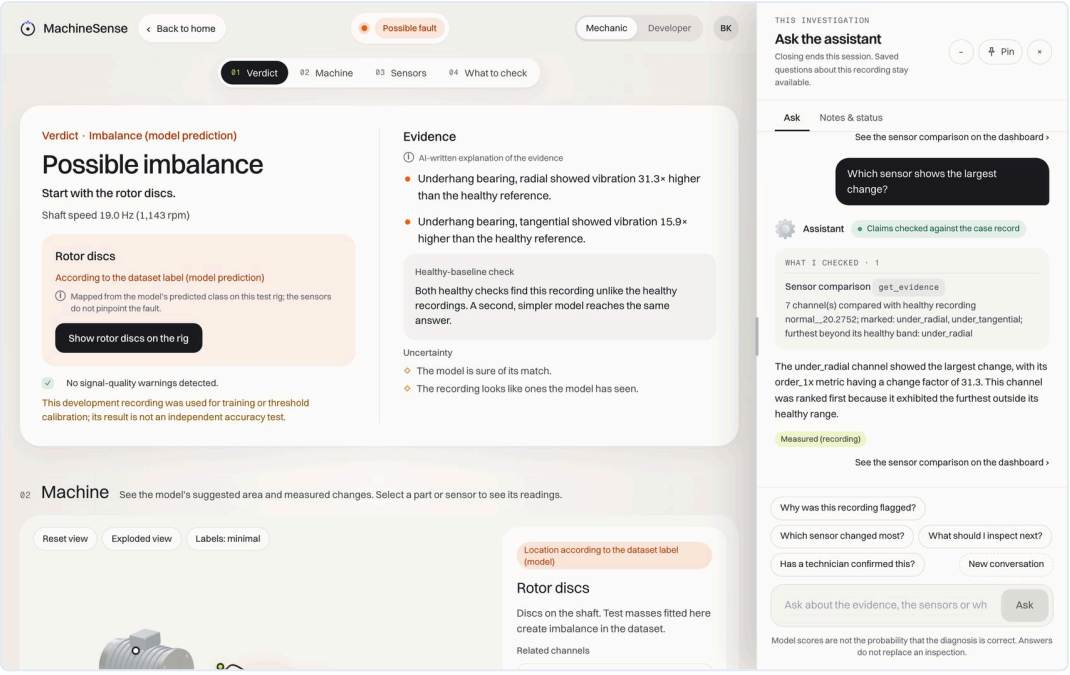
The pipeline page for one case.

5.5 Investigation assistant

Open the assistant panel from a case and type a question about that case, for example:

- Which sensor shows the largest change?
- Why was it flagged?
- What should I check first?

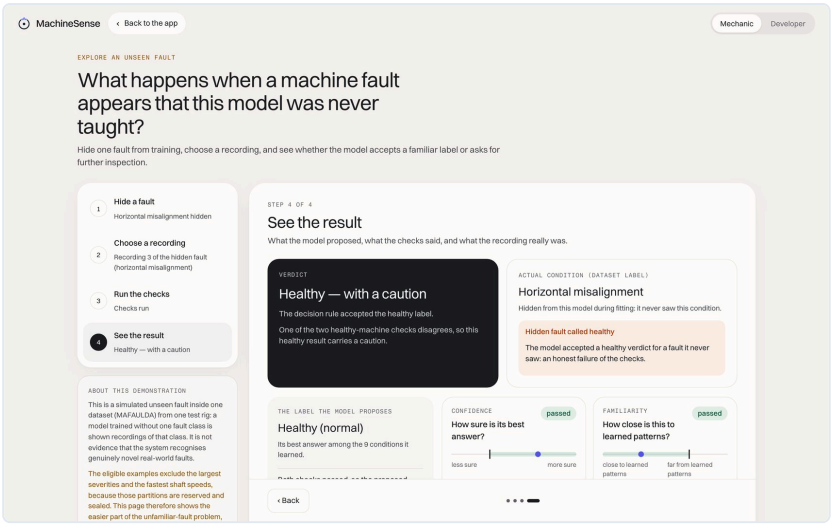
The assistant uses read-only tools on the current case. Code checks every number in its answer before it is shown, and each answer lists its sources. The assistant never changes the verdict. It needs the optional language model (Section 6). Without it, the panel shows the fixed report.



The assistant answering questions about one case, with sources.

5.6 Explore an unseen fault

Choose **Explore an unseen fault** on the home page. Pick one fault class to hide from the model, then run a recording of that class. The page shows what the model did when it had never seen that fault: stopped it for further inspection, gave it a wrong familiar label, or called it healthy. It works from saved results of the hidden-class models.



The unseen-fault page: a hidden fault called healthy.

5.7 Upload your own recording

In the Start investigation dialog, choose upload and select a CSV file in the MAFAULDA format:

- no header row;
- 8 columns in the MAFAULDA channel order (tachometer, three underhang accelerometer axes, three overhang accelerometer axes, microphone);
- 250,000 rows (5 s at 50 kHz), finite values only.

The app explains why any other shape is refused. It also refuses an upload that matches a reserved test recording.

6. Optional: written explanations and the assistant

The app works without a language model. Without one, it shows template explanations, and the assistant shows the fixed report. For AI-written explanations and the investigation assistant:

1. Install LM Studio (lmstudio.ai).
2. Load the model `google/gemma-4-e4b`.
3. Start its local server on port 1234.

The backend finds it at `http://localhost:1234/v1`. No API key is needed, and no data leaves the computer. A written explanation takes about 15 to 20 seconds and appears on the dashboard when ready.

7. Troubleshooting

Problem	What to do
The backend refuses to start with a bundle mismatch	A model file was changed or damaged. Unzip or clone the project again.
The recording list is empty	The backend is not running. Start it (Step 3) and check `/api/health`.
Port 8000 or 5173 is already in use	Stop the other program that uses the port, then start again.
pip downloads very large packages on Linux or Windows	Install the CPU-only PyTorch first (Section 3).
The explanation says "template", or the assistant shows the fixed report	This is expected without LM Studio. Start the local server (Section 6) if you want AI-written text.
A CSV upload is refused	Check the format in Section 5.7. The message names the problem.
A recording that is not in the demo list is requested	The app returns a clear "not available" message. Only the five demo recordings are shipped.

Full reproduction. To retrain the models or run every experiment, download MAFAULDA (kaggle.com/datasets/vuxuancu/mafaulda-full) and follow "Full reproduction" in `SystemCode/HOW_TO_RUN.md`. This is not needed to run or assess the app.